杭州电子科技大学学生考试卷（ A ）卷

考试课程	通信电路与系统	考试日期	2020 年 7 月 3 日	成 绩			
课程号	A040095s	教师号		任课教师姓名	程知群、陈瑾、林弥、周涛、柯华杰、苏江涛、刘艳、刘杰、孙朋飞		
考生姓名		学号（8位）		年级		专业	

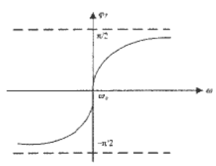
题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题
得分						

Ⅰ. Single Choice (3' each)

1. Which of the following circuits is not part of amplitude modulation transmitting system? (D)

A. Modulator　　B. Main oscillator
C. High frequency power amplifier　　D. Detector

2. The impedance characteristic curve of a series LC resonant circuit is show as below. When the operation frequency is higher than the resonance frequency, the circuit show (A)



A. Inductive　　B. Capacitive
C. Resistor　　D. All of the above

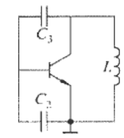
3. When the received signal of the mobile phone is weak，it will enhance the transmitter power to guarantee the communication quantity, which leads to greater electromagnetic radiation to human body. A typical 20 dBm transmitter power of mobile phone corresponds to (C) in mW unit.

A. 1　　B. 10
C. 100　　D. 1000

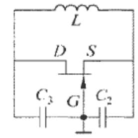
4. Which of the conduction angle (CCA) 2θ is classified as Class C power amplifier? (D)

A. $2\theta=360^0$　　B. $180^0<2\theta<360^0$
C. $2\theta=180^0$　　D. $2\theta<180^0$

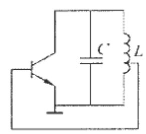
5. By using the phase equilibrium condition, determine which of the oscillation circuits shown below is possible. (D)



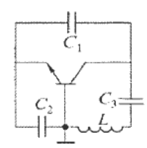
A.



B.



C.

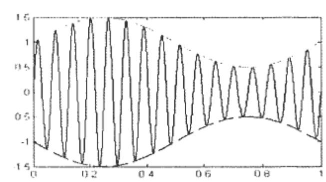


D.

6. In parallel mode crystal oscillator, the crystal works as an inductor, the operating frequency f_0 should satisfy (B)

A. $f_0< f_s$　　B. $f_s <f_0< f_p$
C. $f_0= f_s$　　D. $f_0>f_p$

7. For the standard AM signal shown below, the modulation index m_a is (C)



A. 1 B. 0.2

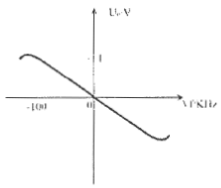
C. 0.5 D. 2

8. In a super-heterodyne radio, if intermediate frequency $f_I = f_L - f_c = 465$kHz, when listening to the radio station with frequency of $f_c = 600$kHz, the interference frequency 1530kHz belongs to (B)

A. Combined frequency interference B. Side-channel interference

C. Cross modulation interference D. Intermodulation interference

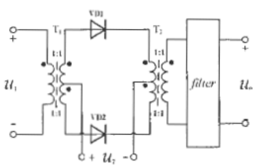
9. The frequency discrimination characteristics curve has been shown below. Known that the output voltage of the frequency discriminator is $u_o(t)=0.5\cos 4\pi\times10^3 t$ (V), the frequency discriminating trans-conductance g_d and maximum frequency shift Δf_m of input signal are (A)



A. -1/100 (V/KHz),50 KHz B. -1/100 (V/KHz),100 KHz

C. -1/200 (V/KHz),50 KHz D. -1/200 (V/KHz),100 KHz

10. The diode multiplier circuit is shown below. Assume the I/V curve characteristics of diode VD1 and VD2 are exactly the same, What type of filter should be used at u_o terminal? (B)



A. Low pass filter B. Band pass filter

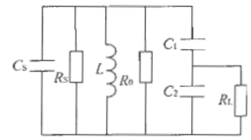
C. Band stop filter D. High pass filter

II. Calculation

1. (8') Capacitor divider impedance conversion circuit is shown below, known that resonant frequency f_0=40 MHz, C_1=20 pF, C_2=20 pF, C_S=10 pF, R_S=20 kΩ, Q_0=100. If R_L and R_S achieves impedance matching

(1) Calculate the access factor n and the value of R_L.

(2) Find out the inductance L, the load quality factor Q_L and bandwidth $B_{0.7}$.



$$n=\frac{C_1}{C_1+C_2}=\frac{20}{20+20}=\frac{1}{2}\quad (1')$$

$$R_L'=\frac{1}{n^2}R_L=(\frac{C_1+C_2}{C_1})^2 R_L=20K\Omega\,,R_l=n^2R_l'=(\frac{C_1}{C_1+C_2})^2R_s=5K\Omega\quad(2')$$

$$C_\Sigma=C_S+\frac{C_1\cdot C_2}{C_1+C_2}=20pF\quad(1')$$

$$f_0=\frac{1}{2\pi\sqrt{LC}}=40MHz\qquad L=\frac{1}{(2\pi f_0)^2C}=0.79\mu H\quad(1')$$

$$R_0=Q_0\omega_0L=19.89K\Omega\qquad(1')$$

$$Q_L=\frac{R_\Sigma}{\omega_0 L}=\frac{R_S//R_0//R_l'}{\omega_0 L}\approx 33.5\quad(1')$$

$$B_{07}=\frac{f_0}{Q_L}=1.19MHz\quad(1')$$

2. (12') A resonant power amplifier with operating at critical point state. Assuming E_C = 24 V, current conduction angle θ = 60°, α_0 (60°) = 0.218, α_1 (60°) = 0.391, collector current pulse maximum value I_{cmax} = 2.2 A, collector voltage utilization factor ξ = 0.9.

(1) Please calculate output power Pout, power of power supply P_{DC}, collector efficiency ηc and collector equivalent load resistor Rc.

(2) If the equivalent load resistor Rc changing suddenly to short. How to change of output power Pout, power supply P_{DC}. Please analyze the reasons of changing and potential damage.

$U_{c1m}=\xi \times E_c = 0.9\times 24 = 21.6V$ (1')

$I_{c1m}=\alpha_1(\theta)\times I_{c\max} = 0.391\times 2.2 \approx 0.86A$ (1')

$P_{out}=\frac{1}{2}I_{c1m}U_{c1m}=\frac{1}{2}\times 21.6\times 0.86\approx 9.29W$ (1')

$I_{c0}=\alpha_0(\theta)\times I_{c\max}=0.218\times 2.2 = 0.48A$ (1')

$P_{DC}=E_c \cdot I_{c0}=24\times 0.48\approx 11.52W$ (1')

$\eta_c=\frac{P_{out}}{P_{DC}}=\frac{9.29}{11.52}\approx 80.6\%$ (1')

$R_c=\frac{U_{c1m}}{I_{c1m}}=\frac{21.6}{0.86}\approx 25.1\Omega$ (1')

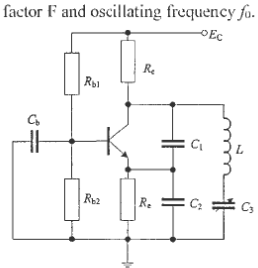
其它条件不变时，集电极等效负载电阻 Rc 短路，功放工作到欠压状态，输出功率也会降低至0；（3'）

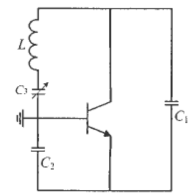
原因是输出电压基波分量会迅速减小到0，虽然集电极电流基波分量会缓慢增加，但二者乘积仍然为0，故输出功率为0。所有的电源功率都用来发热，严重时会烧毁功放管。（2'）

3. (10') The oscillation circuit is shown below, known that Ec=5V, R_{b1}=5.1KΩ, R_{b2}=6.2KΩ, C_b=0.1 uF, C_1=1000 pF, C_2=2000 pF, C_3=60~100 pF, L=50uH

(1) Draw the high frequency AC equivalent circuit, and explain the type of oscillator.

(2) Calculate the feedback factor F and oscillating frequency f_0.





(3')

Clapp circuit (2')

$F=U_f/U_0=C_1/C_1+C_2=1/3$ (2')

$C_\Sigma = C_1 // C_2 // C_3 \approx C_3 = 60\sim100\ pF$ (1')

$$f_0=\frac{1}{2\pi\sqrt{LC_\Sigma}} \quad (1')$$

$$=\frac{1}{2\pi\sqrt{50\times10^{-6}\times(60\sim100)\times10^{-12}}}=2..25\sim2.91\ MHz\ (1')$$

4. (10') A expression of amplitude modulated wave is u=2cos2000πt+0.4cos1800πt+0.4cos2200πt (V). Please answer the following questions:

(1) Find out the power of sideband frequency and the whole power with unit resistor.

(2) Draw the spectrum diagram of the modulated wave and find out the frequency bandwidth B of the modulated wave.

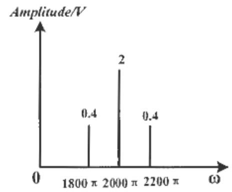
u= 2(1+0.4cos200πt)cos2000πt (V)

$P_{whole}=P_c+P_{sideband}$, for a unit resistor

$P_c=\frac{1}{2}\frac{U_{cm}^2}{R}=0.5\times 2^2=2(W)$ (1')

$P_{sideband}=\frac{1}{2}m_a^2P_c=0.16\ (W)$ (2')

$P_{whole}=P_c+P_{sideband}=2.16\ (W)$ (2')



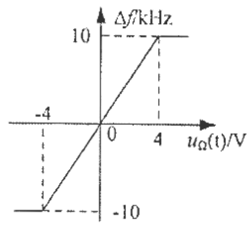
(3')

$$B=2F=2\times\frac{\Omega}{2\pi}=\frac{200\pi}{\pi}=200(\text{Hz})\quad(2')$$

5. (10') The frequency modulation characteristic curve of a FM circuit is shown below. known that the input modulation signal $u_\Omega(t)=2\cos 2\pi\times10^3t$ (V), the carrier signal $u_c(t)=5\cos 2\pi\times10^6t$ (V)

(1) Calculate the frequency sensitivity S, the maximum frequency shift Δf_m, the maximum phase shift m_f.

(2) Write the mathematical expression of the output FM signal $u_{FM}(t)$ and calculate the effective bandwidth B_{FM}.



$$S=\frac{\Delta f}{\Delta U_\Omega}=2.5\ (KHz/V)\quad(2')$$

$$\Delta f_m=SU_{\Omega m}=2.5\times2=5\ (KHz)\quad(2')$$

$$m_f=\frac{\Delta f_m}{F}=5\ (2')$$

$u_{FM}(t)=U_{cm}\times\cos(2\pi\times f_c t+m_f\sin\Omega t)=5\cos(2\pi\times10^6t+5\sin 2\pi\times10^3t)$ (V) (2')

$B_{FM}=2(m_f+1)F=12$ (KHz) (2')

III. Design

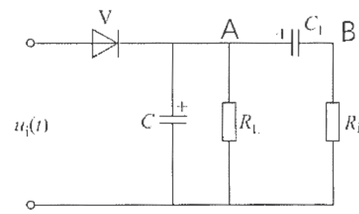
6. (20') The circuit of large signal peak envelope detector is shown below, assuming the detector efficiency η_d is 1.

given $u_i(t)=5\cos2\pi\times465\times10^3t+3\cos2\pi\times10^3t\cos2\pi\times465\times10^3t$ (V), C=0.01μF, C_1=47μF

(1) Find the amplitude modulation index m_a and draw the time domain waveform of $u_i(t)$.

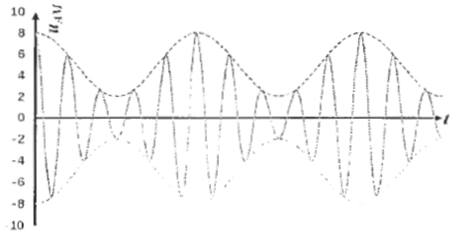
(2) Assure output signal without distortion, please find the maximum load resistance R_L of demodulator, the minimum input resistance Ri.

(3) Write the mathematical expression of $U_A(t)$ and $U_B(t)$.



$u_i(t)=5(1+0.6\cos2\pi\times10^3t)\cos2\pi\times465\times10^3t$ (V) (1')

$m_a=0.6$ (2')



(5')

according to the non-distortion condition ,we get

$$R_L \le \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\Omega C} \quad (2')$$

$$R_L \le \frac{0.8}{0.6} \times \frac{1}{2\pi \times 10^3 \times 0.01 \times 10^{-6}} = 21.2\ (k\Omega)\ (2')$$

according to the non-clipping distortion condition, we have

$$m_a \le \frac{R_i}{R_i + R_L} \quad (2')$$

Put R_L=21.2 kΩ and m_a=0.6 into formula above, we get $R_i \geqslant 31.8$ kΩ (2')

$U_A(t) = 5(1+0.6\cos 2\pi \times 10^3 t)$ (V) (2')

$U_B(t) = 3\cos 2\pi \times 10^3 t$ (V) (2')

座位号：

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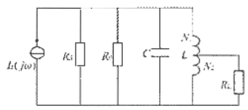
1.(1) what is the name of the circuit shown below? (2 分)
(2) what is mathematical expression of resonant frequency f_0 (2 分)
(3) what is mathematical expression of resonant resistance R_0 (2 分)
(4) what is mathematical expression of Q_0, Q_L (2 分)
(5) what is mathematical expression of bandwidth $B_{0.7}$ (2 分) （合计 10 分）



（1）LC parallel resonant circuit with auto-transformer circuit /Impedance conversion network
加载阻抗变换电路/网络的电感电容并联谐振电路。 (2 分)

(2) $f_0=\frac{1}{2\pi\sqrt{LC}}$。 (2 分)

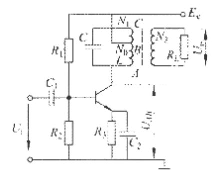
(3) $R_0=\frac{1}{G_0}=\frac{\omega_0^2L^2}{r}=Q_0\omega_0L=\frac{Q_0}{\omega_0C}=Q_0^2r=\frac{L}{Cr}$。 (2 分)

(4) $Q_0=\frac{\omega_0L}{r}=\frac{1}{\omega_0Cr}=\frac{R_0}{\omega_0L}=R_0\omega_0C$

$Q_L=\frac{R_\Sigma}{\omega_0L}=\frac{R_0\,/\!/\,R_s\,/\!/\,R_L'}{\omega_0L}$ $R_L'=\frac{1}{n^2}R_L=\frac{N_1^2}{N_2^2}R_L$。 (2 分)

(5) $B_{0.7}=\frac{f_0}{Q_L}$ (2 分)

2.(1) what is the name of the circuit shown below? (2 分)
(2)what is the function of R_1, R_2, R_3? (1 分)
(3)What is the function of C_1, C_2? (1 分)
(4)What is the function of LC parallel resonant circuits? (1 分)
(5)What is the function of BJT? (1 分)
(6)What is U_i, what is U_o, is U_o smaller that U_i? do they have the same operation frequency/amplitude? (2 分)
(7)Why the R_L partly accessed in the LC parallel resonant circuits? (2 分) （合计 10 分）

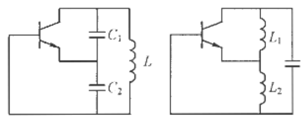


（1）Small signal tuned amplifier （小信号调谐放大器）。 (2 分)
（2）control the operation voltage of transistor （控制晶体管的工作电压/状态）。 (1 分)
（3）coupling capacitor，separate DC，pass wanted high frequency signal （耦合电容，隔阻直流电，通高频有用信号）。 (1 分)
（4）frequency selective，keep wanted HF signal，filter unwanted AC signal （频率选择，保留有用高频信号，过滤无用交变信号）。 (1 分)
（5） amplify signal （放大信号） (1 分)
（6） Ui wanted input signal（输入有用信号）；
Uo wanted output signal（输出有用信号）；
no， Uo is usually bigger than Ui （不， Uo 通常大于 Ui）；
Uo and Ui have same frequency， have different amplitude（Uo Ui 同频不同幅）。 (2 分)
（7） reduce the influence of signal source resistance and load resistance on the circuit（降低电路源端/负载端的阻抗影响） (2 分)

3.A resonant power amplifier with operating at critical point state. Assuming Ec = 5V, current conduction angle (CCA) θ = 60°, α0 (60°) = 0.21, α1 (60°) = 0.39, collector current pulse maximum value Icmax = 0.5A, collector voltage utilization factor ξ =0.8. Please calculate:
(1) output power P_o, (2 分)
(2) DC voltage source power P_{DC}, (2 分)
(3) collector efficiency ηc (2 分)
(4) collector equivalent load resistor R_c,If R_c increases, will the working status change to over-voltage state or under-voltage state? (4 分) （合计 10 分）

(1)collector voltage utilization factor ξ =0.8，$U_{cm}=E_c\cdot\xi$ =4.0(V)，$I_{c1m}=I_{cmax}*\alpha_1(60°)$ =0.5*0.39=0.195(A)
$P_o=0.5*U_{cm}*I_{c1m}$= 0.39(W) (2 分)
(2)$P_{DC}=E_c*I_{c0}=E_c*I_{cmax}*\alpha_0(60°)$ =0.525(W) (2 分)
(3) $\eta c=P_o/P_{DC}$ =74.2% (2 分)
(4) $R_c=\frac{U_{cm}^2}{2P_o}=20.5(\Omega)$ R_c increases lead to the working status change to over-voltage state (4 分)

4.(1) What are the names of the 2 circuits shown below? (2 分)
(2)What is mathematical expressions of oscillation frequency of 2 circuits (3 分)
(3)What is mathematical expression of feedback coefficient of 2 circuits ? (3 分)
(4)The minimum value of necessary amplification gain K_{min} to maintain the oscillation (2 分)
(合计 10 分)



(1)　　　　(2)

(1) capacitance feedback oscillator (Colpitts circuit) and inductance feedback LC oscillator (Hartley circuit) (2 分)

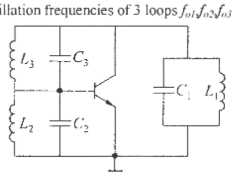
(2)capacitance feedback oscillator : $f_0 \approx \frac{1}{2\pi\sqrt{LC}}$, $C = \frac{C_1C_2}{C_1+C_2}$

inductance feedback LC oscillator: $f_0 \approx \frac{1}{2\pi\sqrt{LC}}$, $L = L_1 + L_2 + 2M$, M is mutual inductance (3 分)

(3) capacitance feedback oscillator : $F \approx \frac{C_1}{C_2}$ inductance feedback LC oscillator: $F \approx \frac{L_2+M}{L_1+M}$ (3 分)

(4)to maintain the oscillation, oscillators should Balance condition of oscillation: K*F=1, K_{min}=1/F
capacitance feedback oscillator : $K_{min} = C_2 / C_1$
inductance feedback LC oscillator: $K_{min} = \frac{L_1+M}{L_2+M}$ (2 分)

5.A high equivalent circuit is shown below, assuming the following six conditions:
1)$L_1C_1>L_2C_2>L_3C_3$,2)$L_1C_1<L_2C_2<L_3C_3$,3)$L_1C_1=L_2C_2=L_3C_3$,
4)$L_1C_1=L_2C_2>L_3C_3$,5)$L_1C_1<L_2C_2=L_3C_3$,6)$L_1C_1>L_3C_3>L_2C_2$
(1)please select which kinds of conditions above likely to oscillate (3 分)
(2)corresponding to oscillation conditions in (1), what is the name (type) of oscillation circuit (2 分) ,
(3)mathematical expressions of natural oscillation frequencies of 3 loops f_{o1},f_{o2},f_{o3} (2 分)
(4) the relationship between the oscillation frequency of the whole circuit f_{osc} and the natural oscillation frequencies of 3 loops f_{o1},f_{o2},f_{o3} ? (3 分) (合计 10 分)

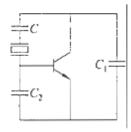


(1) 1), 2) and 4) conditions likely to oscillate (3 分)

(2) $f_{o1} = \frac{1}{2\pi\sqrt{L_1C_1}}$, $f_{o2} = \frac{1}{2\pi\sqrt{L_2C_2}}$, $f_{o3} = \frac{1}{2\pi\sqrt{L_3C_3}}$, (2 分)

(3) capacitance feedback oscillator 1), 4)
inductance feedback oscillator 2) (2 分)
(4) For capacitance feedback oscillator 1), 4) : $f_{o1} \leqslant f_{o2} < f_{osc} < f_{o3}$
For inductance feedback oscillator 2): $f_{o1}, f_{o2} > f_{osc} > f_{o3}$ (3 分)

6.(1) which kind of oscillator is the circuit shown below? (2 分)
(2)what is the function of crystal? It works in series mode or parallel mode (3 分)
(3) please draw the equivalent circuit of the crystal (2 分)
(4) mathematical expressions of the oscillation frequency of the circuit(3 分)
(合计 10 分)



(3)  (2 分)

(1)Crystal oscillator (2 分)
(2)the crystal works as an inductor, works in parallel mode (3 分)

(4) $$\frac{1}{C_\Sigma} = \frac{1}{C_1} + \frac{1}{C_2 + \cfrac{1}{\cfrac{1}{C} + \cfrac{1}{C_1} + \cfrac{1}{C_1}}},\qquad \frac{1}{C_\Sigma} \approx \frac{1}{C_q} + \frac{1}{C_0 + C} = \frac{C_q(C_0+C)}{C_q+C_0+C}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_q \cdot \frac{C_q(C_0+C)}{C_q+C_0+C}}}$$ (3 分)

7.Given 2 signal voltage u1 and u2, the corresponding frequency part are:
u_1=100cos1000πt+20cos800πt+20cos1200πt(V)

u_2=20cos800πt+20cos1200πt(V)

Please find:
(1)Which modulated waves are u_1 and u_2? (2 分)
(2)The mathematical expressions of modulated waves of u_1 and u_2. (3 分)
(3) The power of side-band and the whole power with resistor 50 Ohms. (3 分)
(4) The frequency bandwidth B of the modulated wave. (2 分) (合计 10 分)

座位号：　　　学号：　　　课程号：　　　教师号：

.(1) u_1 is a normal AM wave, u_2 is a DSB AM wave with suppressed carrier (2 分)

(2) u_1=100cos1000πt+20cos800πt+20cos1200πt(V) =100(1+0.2cos200πt)cos1000πt ,

u_2=20cos800πt+20cos1200πt = 40cos200πtcos1000πt, (3 分)

(3) For u1, $P_{whole}=P_c+P_{sideband}$, for a 50 Ohms resistor

$P_c=\frac{1}{2}\frac{U_{cm}^2}{R}=0.5\times100^2\div50=100(W)$, $P_{sideband}=\frac{1}{2}m_a^2P_c=0.5\times0.2^2\times100=2(W)$, $P_{whole}=P_c+P_{sideband}=102(W)$

For u2, $P_{whole}=P_{sideband}=8(W)$ (3 分)

(4) For u1, frequency bandwidth $B_1=2F=2\times\frac{\Omega}{2\pi}=\frac{200\pi}{\pi}=200(Hz)$

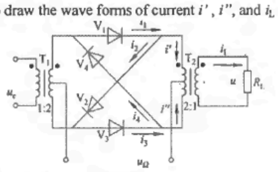
For u2 frequency bandwidth $B2=2F=2\times\frac{\Omega}{2\pi}=\frac{200\pi}{\pi}=200(Hz)$ (2 分)

8.(1) what is the function of the circuit shown below? please analysis it's working mechanism (4 分)

(2)Draw the equivalent circuit when u_c positive half cycle and negative half cycle (4 分)

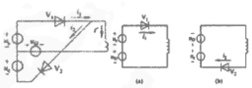
(3)Give the mathematical expressions of current i' and i'' when uc positive half cycle and negative half cycle, and the i_L through load resistor R_L(3 分)

(4) draw the wave forms of current i', i'', and i_L (3 分) (合计 14 分)

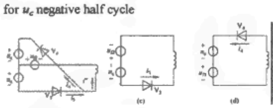


(1) named：Diode ring amplitude modulated circuit， used to produce amplitude modulated signal with suppressed carrier， u_c positive half cycle.V1,V2 on，V3,V4 off；negative half period,V3,V4 on，V1,V2 off (4 分)

(2) for u_c positive half cycle



for u_c negative half cycle



(4 分)

(3) for u_c positive half cycle

$i_1=gk_1(\omega_c t)(u_c+u_\Omega)$ $i_2=gk_1(\omega_c t)(u_c-u_\Omega)$ $i'=i_1-i_2=2gk_1(\omega_c t)u_\Omega$

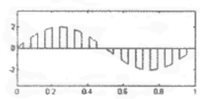
for u_c negative half cycle

$i_3=-gk_1(\omega_c t+\pi)(u_c-u_\Omega)$ $i_4=-gk_1(\omega_c t+\pi)(u_c+u_\Omega)$ $i''=i_3-i_4=-2gk_1(\omega_c t+\pi)u_\Omega$

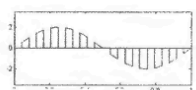
$i_L=i'-i''=2gu_\Omega[k_1(\omega_c t)-k_1(\omega_c t+\pi)]=2gu_\Omega k_2(\omega_c t)$

$=2gU_{\Omega m}\cos\Omega t\cdot\frac{4}{\pi}[\cos\omega_c t-\frac{1}{3}\cos3\omega_c t+\frac{1}{5}\cos5\omega_c t+\Lambda]$ (3 分)

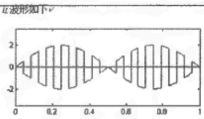
(4) i' 波形如下



i''波形如下



i_L波形如下



(3 分)

9.We have a modulation signal, $u_\Omega(t)=U_{\Omega m}\cos\Omega t$

and carrier wave signal $u_c(t)=U_{cm}\cos\omega_c t$

and their wave forms are listed in figure below, Please draw the modulated waveform:

(1) u_{AM} standard amplitude modulation (AM) wave form (2 分)

(2) $u_{DSB\text{-}AM}$ double side band AM wave form (2 分)

(3) $u_{SSB\text{-}AM}$ single side band AM wave form (2 分)

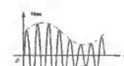
(4) u_{FM} frequency modulation (FM) wave form (2 分)

(5) u_{PM} phase modulation wave form (1 分)

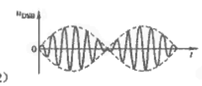
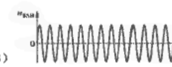
(6) Which modulation method can work in both AM and FM (1 分)

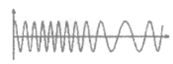
Which modulation method can work in both AM and PM (1 分) (合计 11 分)

(1)



(2 分)

(2)  (2 分) (3)  (2 分)

(4)  (2 分) (5)  (1 分)

(6) Double side-band (DSB) modulation method can work in both AM and PM (1 分)

Single side-band (SSB) modulation method can work in both AM and FM (1 分)

10.Consider the characteristics of non-linear devices and non-linear circuits, from the perspective of environmental protection and sustainable development, please:

(1) describe the possible harm/pollution of high frequency circuit/system (3 分)

(2) list 1-2 method to avoid the harm/pollution of high frequency circuits (2 分)（合计 5 分）

（1）电台、通信基站、通信终端、大功率交变电器的启动/关停瞬间的电磁污染等等 (3 分)

（2）对个人来说：可以穿导电性能良好的屏蔽服等等。对电器来说，可以做好电磁兼容/电磁干扰，避免污染环境。(2 分)